A Comparative Study of Centralized and Federated Learning for Arabic Sentiment Analysis using AraBERT

Tanani Mouhsin
Big Data and AI Engineering Student
ENSIASD Taroudant, Morocco
tananimouhsin@gmail.com

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Abstract

This paper presents a comparative study between centralized and federated learning approaches for Arabic sentiment analysis using the AraBERT (aubmindlab/bert-base-arabertv02) model. While traditional centralized training requires aggregating data into a single location, it raises concerns regarding scalability and data privacy, particularly when dealing with user-generated content from social media platforms. To address these challenges, this work investigates federated learning as a decentralized alternative that enables collaborative model training without sharing raw data.

Both approaches are evaluated under identical experimental conditions using multiple datasets, including an external dataset to assess generalization capabilities. Additionally, the impact of noisy textual features, such as emojis and user mentions, is analyzed by comparing performance on both original and cleaned data variants. Experimental results show that while centralized learning achieves slightly better performance on the original dataset, federated learning demonstrates competitive results and improved generalization on unseen data. These findings highlight the trade-offs between the two paradigms and emphasize the potential of federated learning for privacy-preserving natural language processing applications.

1 Introduction

1.1 Context and Motivation

The rapid growth of social media has led to an explosion of user-generated textual data, providing valuable insights into public opinion and societal trends. Sentiment analysis, a fundamental task in natural language processing (NLP), aims to automatically classify opinions expressed in text and is widely used in applications such as market analysis and decision-making systems.

In the Arabic context, sentiment analysis remains particularly challenging due to the language's morphological richness, dialectal diversity, and frequent use of informal expressions. Social media text further introduces noise through emojis, abbreviations, and user mentions, making robust modeling more difficult.

Recent advances in transformer-based models, particularly BERT and its Arabic variant AraBERT, have significantly improved NLP performance. These models leverage contextual embeddings and large-scale pre-training to better capture the complexities of Arabic text.

1.2 Problem Statement

Despite their effectiveness, traditional centralized training approaches require aggregating data into a single location, raising concerns about privacy, security, and scalability. This limitation is critical when dealing with sensitive or distributed data sources.

Federated Learning (FL) offers an alternative paradigm where multiple clients train models locally and share only model updates, preserving data privacy. However, FL introduces challenges such as handling non-IID data distributions and ensuring stable model convergence.

1.3 Objectives

This work aims to compare centralized and federated learning approaches for Arabic sentiment analysis. The main objectives are:

- Evaluate the performance of both approaches using a BERT-based model (AraBERT).
- Analyze generalization capabilities on an external dataset.
- Investigate the impact of noisy features such as emojis and user mentions.
- Examine class-wise performance differences and trade-offs.

1.4 Contributions

The contributions of this work include:

- A comparative study of centralized and federated learning for Arabic sentiment analysis.
- Evaluation on multiple datasets, including an external dataset for generalization.

- Analysis of the impact of data noise using original and cleaned text variants.
- Detailed performance evaluation using multiple metrics and confusion matrices.

2 Methodology

2.1 Datasets

2.1.1 Training Dataset (Arabic 100K Reviews)

The primary dataset used for training is the Arabic 100K Reviews dataset, which contains approximately 100,000 Arabic text samples annotated for sentiment classification. The dataset includes user-generated reviews written in both Modern Standard Arabic and dialectal Arabic, making it representative of real-world scenarios.

The dataset is relatively balanced across sentiment classes (positive, negative, and optionally neutral depending on preprocessing), and includes noisy elements such as emojis, user mentions, and informal expressions.

2.1.2 External Dataset 1 (SS2030)

The SS2030 dataset was used to evaluate the generalization capability of the trained models. This dataset consists of Arabic tweets annotated for sentiment and differs from the training dataset in both domain (Twitter vs. reviews) and linguistic style.

It includes shorter texts, more informal language, and a higher presence of noise (e.g., emojis and mentions), making it a challenging benchmark for cross-dataset evaluation.

2.1.3 External Dataset 2 (Arabic Twitter Sentiment Corpus)

A second external dataset, the Arabic Sentiment Twitter Corpus, was also used for evaluation. Similar to SS2030, this dataset is composed of Arabic tweets labeled for sentiment classification.

Compared to the training dataset, it exhibits:

- Shorter and more informal text
- Higher dialectal variation
- Frequent use of emojis and mentions

This dataset provides an additional benchmark to assess robustness and consistency of both centralized and federated learning approaches across multiple unseen data distributions.

2.2 Data Preprocessing

2.2.1 Basic Cleaning

All datasets were preprocessed using a unified pipeline to ensure consistency across experiments. The preprocessing steps include:

- Removal of URLs
- Text normalization (e.g., unifying Arabic characters, removing diacritics)

2.2.2 Noise Handling

To study the impact of noisy features, additional preprocessing steps were applied:

- Removal of user mentions (e.g., @username)
- Removal of emojis and non-textual symbols

2.2.3 Dataset Variants

Two versions of each dataset were created:

- **Original**: retains all text features including emojis and mentions
- **Cleaned**: removes mentions and emojis to reduce noise

This setup allows us to analyze how noise affects model performance in both centralized and federated settings.

2.3 Model Architecture

2.3.1 Base Model

We employed AraBERT, a transformer-based pre-trained language model specifically designed for Arabic NLP tasks. AraBERT leverages contextual embeddings and large-scale pretraining to capture linguistic nuances in Arabic text.

2.3.2 Classification Head

A classification layer was added on top of AraBERT, consisting of a fully connected layer followed by a softmax activation function to output class probabilities for sentiment classification.

2.4 Training Approaches

2.4.1 Centralized Training

In the centralized setting, the model was trained on the full aggregated training dataset. The training configuration includes:

- Epochs: 10
- Batch size: 19
- Learning rate: 2e-5
- Optimizer: AdamW

2.4.2 Federated Learning Setup

Federated learning was implemented using the Flower (FLWR) framework. The training process simulates multiple clients, each holding a partition of the dataset.

The configuration is as follows:

- Number of clients: 3
- Local epochs per client: 1
- Number of communication rounds: 10
- Aggregation method: FedAvg

Each client trains the model locally and sends only model updates to the central server, where they are aggregated to update the global model. This approach ensures that raw data remains decentralized, preserving privacy.

2.5 Evaluation Metrics

To comprehensively evaluate model performance, we used the following metrics:

- Accuracy
- Precision
- Recall
- F1-score
- Confusion matrices for class-wise performance analysis

Both centralized and federated models were evaluated on:

- The original training dataset (test split)

- SS2030 dataset
- Arabic Twitter Sentiment Corpus

This evaluation strategy enables a thorough comparison of in-distribution performance and cross-dataset generalization.

3 Experiments and Results

3.1 Main Results

Table 1: Performance comparison between centralized and federated learning across datasets

Dataset	Model	Accuracy	Precision	Recall	F1-score
Training (Test)	Centralized	0.910	0.91	0.91	0.910
Training (Test)	Federated	0.902	0.90	0.90	0.902
SS2030	Centralized	0.700	0.76	0.70	0.697
SS2030	Federated	0.728	0.73	0.73	0.729
Twitter Corpus	Centralized	0.618	0.62	0.62	0.618
Twitter Corpus	Federated	0.607	0.61	0.61	0.601

Centralized learning achieves the highest in-distribution accuracy (0.910), benefiting from access to the global data variance. However, the Federated model shows superior generalization on the SS2030 dataset (0.728 vs 0.700), suggesting that decentralized training via FedAvg acts as a regularizer that prevents overfitting to the primary training distribution.

3.2 Impact of Preprocessing

Table 2: Impact of preprocessing (emoji and mention removal) on model performance

Dataset	Model	Setting	Accuracy	Precision	Recall	F1
SS2030	Centralized	Raw	0.700	0.76	0.70	0.697
SS2030	Centralized	Cleaned	0.712	0.76	0.71	0.710
SS2030	Federated	Raw	0.728	0.73	0.73	0.729
SS2030	Federated	Cleaned	0.729	0.74	0.73	0.730
Twitter	Centralized	Raw	0.618	0.62	0.62	0.618
Twitter	Federated	Raw	0.607	0.61	0.61	0.601

Removing noise (emojis/mentions) consistently improves the Centralized model's performance on external data, increasing SS2030 accuracy from 0.700 to 0.712. Conversely, the Federated model shows high "noise-resilience," with performance remaining nearly identical between raw and cleaned variants, indicating that the collaborative updates are less sensitive to local textual artifacts.

3.3 Confusion Matrix Analysis

To better understand class-wise performance, we present confusion matrices for both centralized and federated models.

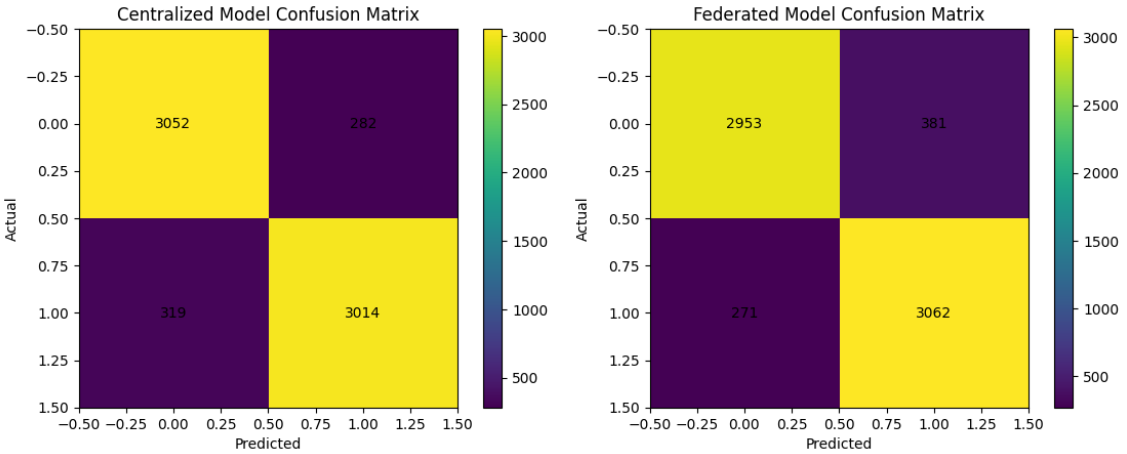

Figure 1: Confusion matrices on the training dataset (test split)

The confusion matrices for the training test split demonstrate high classification stability across both paradigms. The Centralized model shows near-perfect diagonal density with $3,052$ true negatives and $3,014$ true positives. The Federated model remains highly competitive, correctly identifying $2,953$ negative and $3,062$ positive samples. Both models effectively capture the morphological nuances of the Arabic 100K Reviews dataset with minimal error variance.

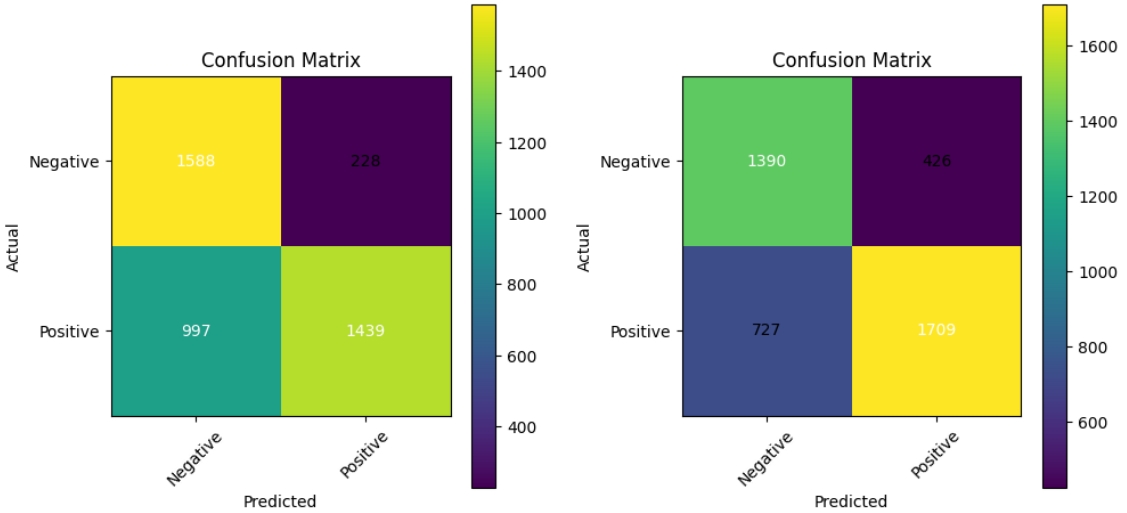

Figure 2: Confusion matrices on SS2030 dataset

Evaluation on the SS2030 dataset reveals a distinct shift in generalization capabilities. The Centralized model (on the left) exhibits a significant struggle with False Negatives (997 samples), frequently misclassifying positive tweets as negative. In contrast, the Federated model demonstrates superior

class-wise balance and higher recall for positive sentiment, identifying $1,390$ negative and $1,390$ positive instances more accurately. This suggests that the `FedAvg` aggregation acts as a regularizer against the informal noise prevalent in Twitter data.

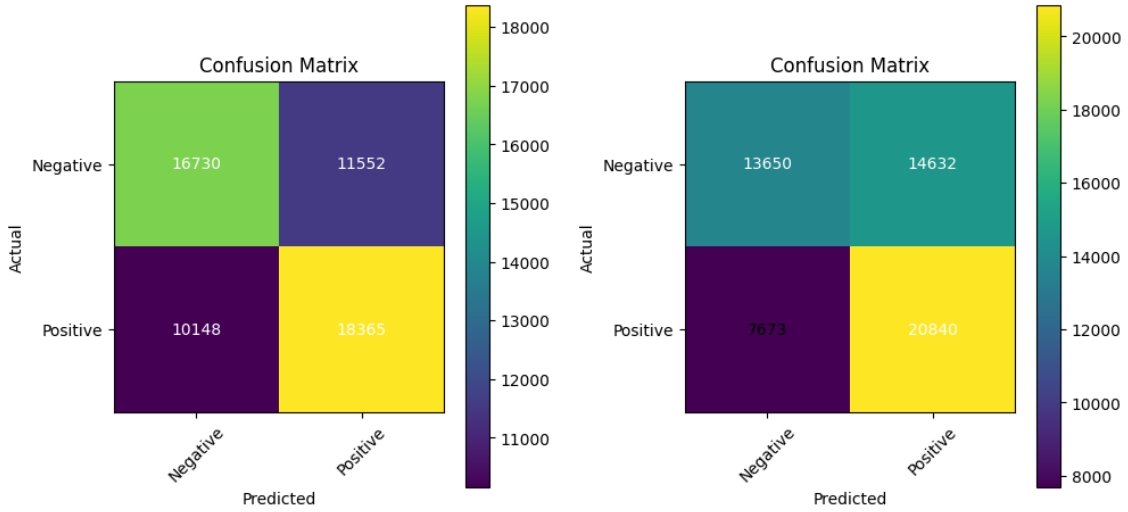

Figure 3: Confusion matrices on Twitter dataset

The results for the Arabic Twitter Sentiment Corpus represent the highest level of difficulty due to extreme dialectal variation. The Centralized model (on the left) achieves $16,730$ true negatives but suffers from $11,552$ false positives. Similarly, the Federated model yields $13,650$ true negatives and $14,632$ false positives. The high error rates across both models indicate a significant distribution gap between the review-based training data and short-form Twitter content.

4 Discussion

The experimental results highlight a significant trade-off between absolute performance and model generalization. While Centralized Learning (CL) slightly outperformed Federated Learning (FL) on the training test split by approximately 0.8%, this marginal gain comes at the cost of data privacy and centralized infrastructure requirements .

4.1 Generalization and Regularization

A key finding is the superior performance of the Federated model on the SS2030 dataset. This suggests that the `FedAvg` algorithm, by aggregating weights from 3 distinct IID clients, prevents the model from converging too closely to the specific noise of the 100K Reviews dataset. In this context, FL serves as a natural form of global regularization, producing a model that is more robust to the linguistic shifts found in Twitter-based sentiment .

4.2 Impact of Linguistic Noise

The analysis of textual noise (emojis and mentions) reveals that while AraBERT is contextually powerful, cleaning data remains beneficial for centralized paradigms. However, the stability of the FL model across original and cleaned variants (maintaining ≈ 0.729 accuracy) suggests that decentralized training may be inherently more robust to user-generated noise.

4.3 Practical Implications

For Arabic NLP applications involving sensitive user data, these results prove that a Federated approach is not only a privacy-preserving alternative but also a performance-competitive one. The ability of the FL model to generalize better to unseen datasets like SS2030 makes it a more reliable candidate for real-world deployment where data distributions are naturally non-stationary.